Alexandra Neagu

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Hi, I am Alexandra — a second-year PhD researcher at Imperial College London studying how we can apply Large Language Models (LLMs) to support self-study in STEM, with publications at top AI+Education venues (ICML Workshop 2026, ACM L@S 2026, EERN 2025). I worked for nine months at AWS Professional Services shipping serverless ML systems for internal and external customers.

Education

PhD in Applied AI in Education – Imperial College London *2024 – 2028*

- Exploring student-LLM interactions through mixed-methods analysis at scale (discourse analysis, LLM-as-judge, pattern clustering); supervised by Dr Peter B. Johnson & Dr Rhodri Nelson
- Designing and evaluating LLM-powered dialogic feedback on Lambda Feedback, a self-study platform used across all STEM faculties at Imperial

MEng Electronic and Information Engineering – Imperial College London *2020 – 2024*

- First Class Honours; Imperial College President's Undergraduate Scholarship for academic excellence.
- **Final Year Project:** *"Enhancing Puzzle-Solving with User-Tailored AI Assistance" (supervised by Dr Nicole Salomons)* — controlled user study (n=25) measuring whether skill-tailored LLM hints improve novice learning on Nonogram puzzles. *Finding:* LLM assistance improved task efficiency but not learning gains.
- Relevant modules: Human-Centred Robotics, Self-Organised Multi-Agent Systems, Machine Learning, Privacy Engineering (Anonymisation, Differential Privacy), Computer Vision and Robotics (SLAM).

International Baccalaureate Diploma – Sevenoaks School, Kent *2018 – 2020*

- HL: Mathematics (7), Physics (7), Geography (6)   SL: English Literature (5), German B (5), Mandarin B (6).

Publications

Rethinking Scaffolding in LLM Tutors: The Interactional Mismatch Between Benchmarks and Real-World Deployments *ICML Pluralistic Alignment Workshop 2026*

***Neagu, A.**, Wong, JTH., Messer, M., Johnson, P.B. and Nelson, R.*

- Educational LLM chatbots are aligned and evaluated assuming students take up their scaffolding. We reveal that real-world interactions are much more diverse and are highly driven by the subjective learning context and values of each student. Thus, we propose recommendations for future design and evaluation of educational chatbots.

'How Do I...?': Procedural Questions Predominate Student-LLM Chatbot Conversations *arXiv preprint, 2026*

***Neagu, A.**, Messer, M., Johnson, P.B. and Nelson, R.*

- Student questions drive the prompts sent to educational chatbots, and so shape how pedagogically useful the responses are. We set up a multi-LLM rater analysis to apply existing question-classification schemas. We show that procedural "how-to" questions predominate in 2 distinct learning contexts, even more so under summative pressure.

μEd API: Towards a Shared API for Education Microservices *ACM Learning@Scale 2026*

*Sölch, M., **Neagu, A.**, Messer, M., Johnson, P., Kortemeyer, G., Ng, S.S.H., Lim, F.S. and Krusche, S.*

- We propose μEd, an initial platform-independent API for educational microservices to automatically provide feedback, assessment, or chatbots, built from systems already in use at four institutions that are now adopting it.

Chatbots for Dialogic Feedback during Self-Study: The Importance of Contextual Information *EERN Symposium 2025*

***Neagu, A.**, Johnson, P.B. and Nelson, R.*

- The context given to an educational LLM chatbot shapes how the interaction unfolds. Across 23 conversations, students focused on their impasse and simply referenced contextual information. We argue that context-aware chatbots can lower exogenous cognitive load, if configured to preserve student agency and productive struggle.

Workshops & Tutorials

Tutorial (half-day)	**Build and Deploy Microservices for Automated Feedback**. Developing algorithms to provide automated feedback to student submissions. *With Johnson P.B., Messer M., Lim F.S.*	*Festival of Learning 2026*
Workshop (full-day)	**PEAF 2026: Pedagogical Evaluation of Automated Feedback**. *With Messer M., Johnson P.B., Kandiko C., Savelka J., Woodhead S.*	*Festival of Learning 2026*

Research Experience

SEFI Summer School on Engineering Education Research – Aalborg University, Denmark *May 2026*

- Selected for a 1-week course hosted by the Centre for Problem-Based Learning in Engineering Science and Sustainability
- Adopting a critical mindset and mixed method methodologies for interdisciplinary education research.

LearnLab Summer School – Carnegie Mellon University, USA *Aug 2025*

- Selected for a 1-week course on technology-enhanced learning and intelligent tutoring systems (ITS).
- Trained a simulated student under the Knowledge-Learning-Instruction (KLI) framework to acquire politeness as knowledge components via interaction with an ITS tutor in a Chinese-learning task.

Industry Experience

Professional Services Industrial Placement – Amazon Web Services, UK
Apr – Sept 2023

- **Text-to-Speech pipeline proposal on AWS Polly (Emirates iXR use case):** designed and prototyped an end-to-end serverless TTS pipeline to generate audio for 3D avatars in employee training scenarios; packaged as a reusable Terraform deployment with a flexible UI, with resiliency, security, and high availability as core design constraints.
- **Global Monitoring & Alerting for the global EC2 fleet of AWS's virtual-desktop service**: built the back-end automation, QuickSight dashboards, and alarming system tracking metrics like active users, time-on-instance, CPU anomalies, idle instances — enabling the service team to right-size the fleet and flag unused instances or suspicious activity.
- Presented technical decisions in architecture reviews with senior consultants across EMEA and the US.

Professional Services Summer Intern – Amazon Web Services, UK
Jul – Sept 2022

- Migrated AWS's internal Excel-based Landing Zone Assessment onto the AWS Assessment Tool platform; shift allowed the offering from internal-only to customer- and partner-facing, enabling broader adoption as a standard practice.
- Cut consultant assessment time by ∼**20%** (∼80h workflow) via automated report generation; scoped and requested platform features (e.g. weighted scoring) and implemented custom UI components under the new tool's constraints.

Skills & Certifications

Human-Centred ML & AI: Large Language Models, prompt engineering, LLM-as-judge & human evaluation, qualitative analysis at scale, user studies, PyTorch, agentic workflows.

Programming: Python, TypeScript, SQL, GraphQL, C#, Bash, Git, Unity.

Cloud & MLOps: AWS (Certified Machine Learning Speciality, Solutions Architect Associate, Cloud Practitioner), Google Cloud Platform, Docker, Pulumi, Terraform.

Languages: English (native), Romanian (native), German (B1-B2), Mandarin (HSK 4-5), French (A1-A2).

Selected Projects *(see more on my personal page)*

Avatour – VR-mediated Robot Telepresence [University Project]
Jan – Mar 2024

- **Outcome:** ran a controlled user study across 2D and VR interfaces on operating a Boston Dynamics SPOT robot for exploring restricted real-world locations; findings contradicted our hypothesis: prior user experience, not modality, dominated trust and immersion with the teleoperation interface.
- **Skills:** user study design, mixed-methods HCI evaluation, human-centered interface design.

CharIOT – Cleanroom Monitoring System [University Project]
Jan – Mar 2023

- **Outcome:** distributed IoT system enabling cleanroom researchers to detect environmental failures (temperature, pressure, humidity, particulates) in real time, with user-defined alarm thresholds synchronised across devices.
- **Skills:** distributed systems design, real-time sensor data pipelines, cloud-edge integration.

ZeroToOne – AR/VR Circuit Design Simulator [Hackathon]
Feb 2022

- **Outcome:** an AR/VR Android app delivering step-by-step tutor flow for remote hardware lab circuit assembly.
- **Skills:** drove idea-to-MVP process, prototyping under time & cost constraints, AR/VR interaction design for novice users.

External Courses

- Foundations of Engineering Education, University at Buffalo, NY (online, 12 weeks). *2025*
- Nvidia: Accelerating CUDA C++ applications with multiple GPUs (2 days). *2024*
- Computer Vision Course, Machine Learning University on AWS SageMaker (online, 3 days). *2023*
- AWS Certified SysOps Administrator Associate, Udemy (online, 30 hrs). *2023*

Interests

Vice President Industry, EESoc – Imperial College London
2021 – 2023

- Built partnerships between companies and the Electrical Engineering Society; led the organisation of one of Imperial's largest student-led Careers Fairs and Intro-to-Industry Events.

Hackathon Projects
ongoing

- Participated or volunteered at hackathons every year since 2021, building projects across various domains and technologies. (ICHack 2021-2026, AWS Interns Hack 2023, WSET Hack-her-thon 2021-2022).

Travel & Cross-Cultural Exchange
ongoing

- Travel across Europe, USA, China, Peru, India, Japan, Madagascar, Costa Rica, Tanzania, and Egypt — a core part of who I am: I love experiencing different cultures and the discrepancies in how people approach life.